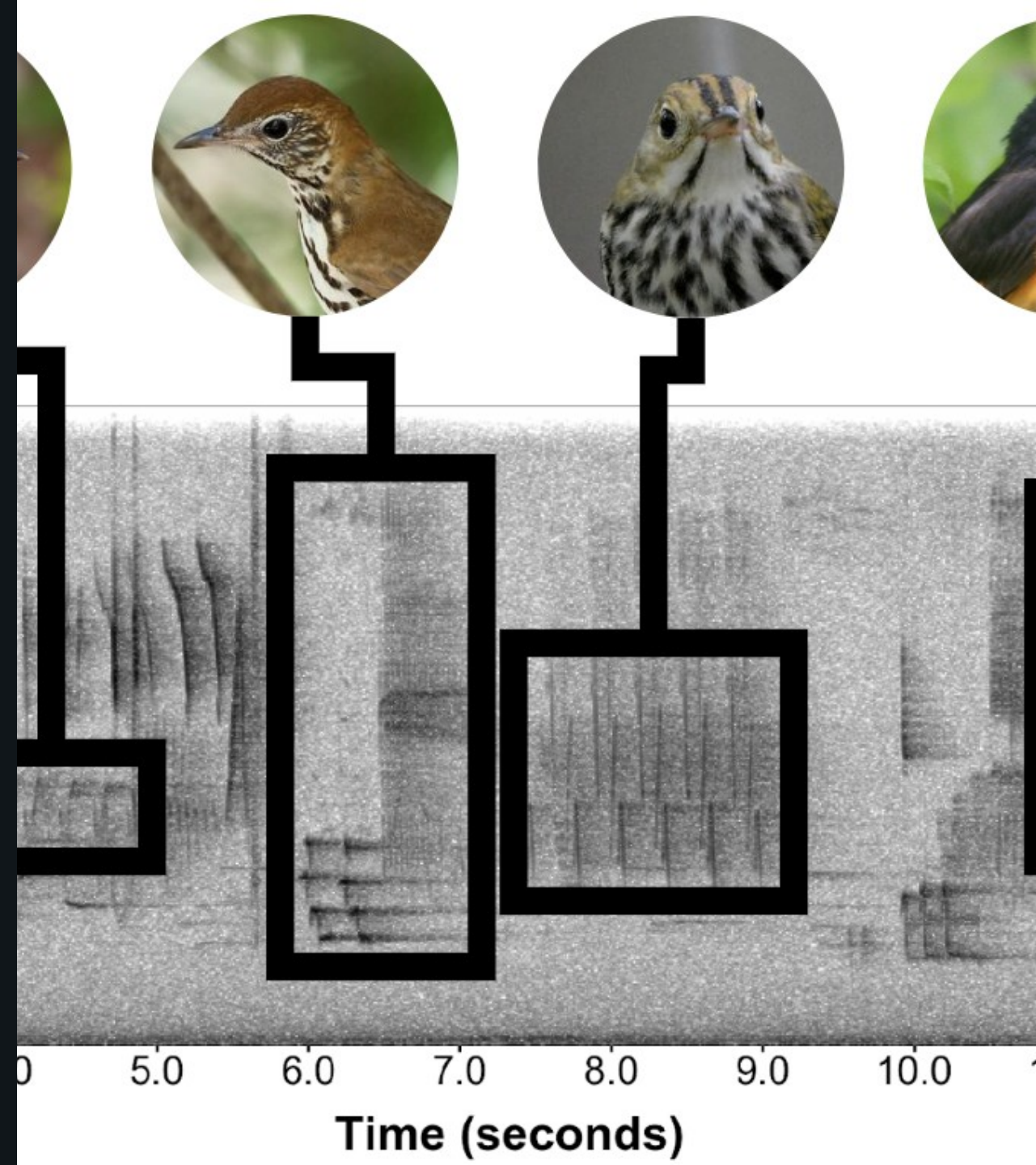
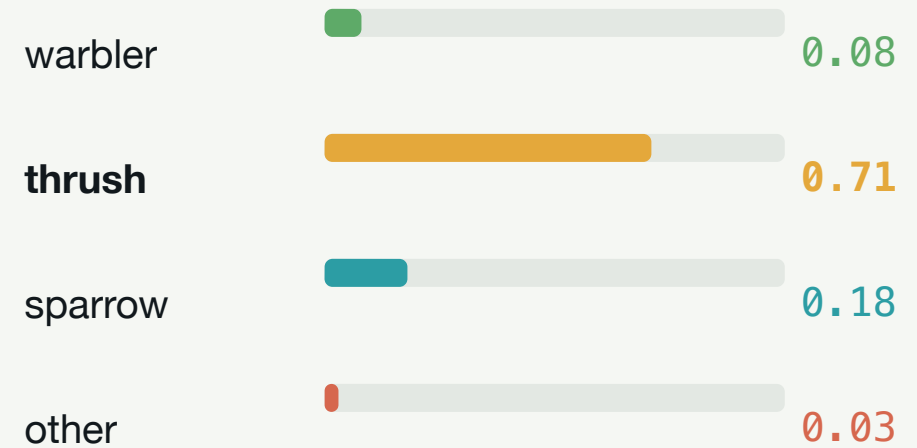
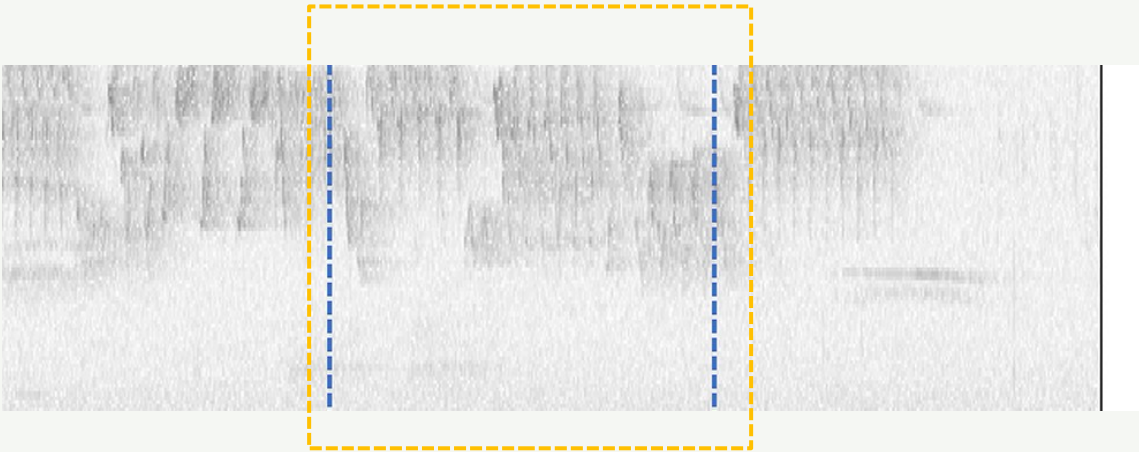


What counts as evidence?

Evaluating a classifier



Model Predictions



prediction: thrush

But was the bird actually a thrush?

Single Examples and Reliability

Inference 1

true: warbler

predicted: warbler

Inference 2

true: thrush

predicted: thrush

Inference 3

true: sparrow

predicted: sparrow

What can we actually conclude from this?

Demonstration and Evaluation

DEMONSTRATION

**Did the system work
on these examples?**

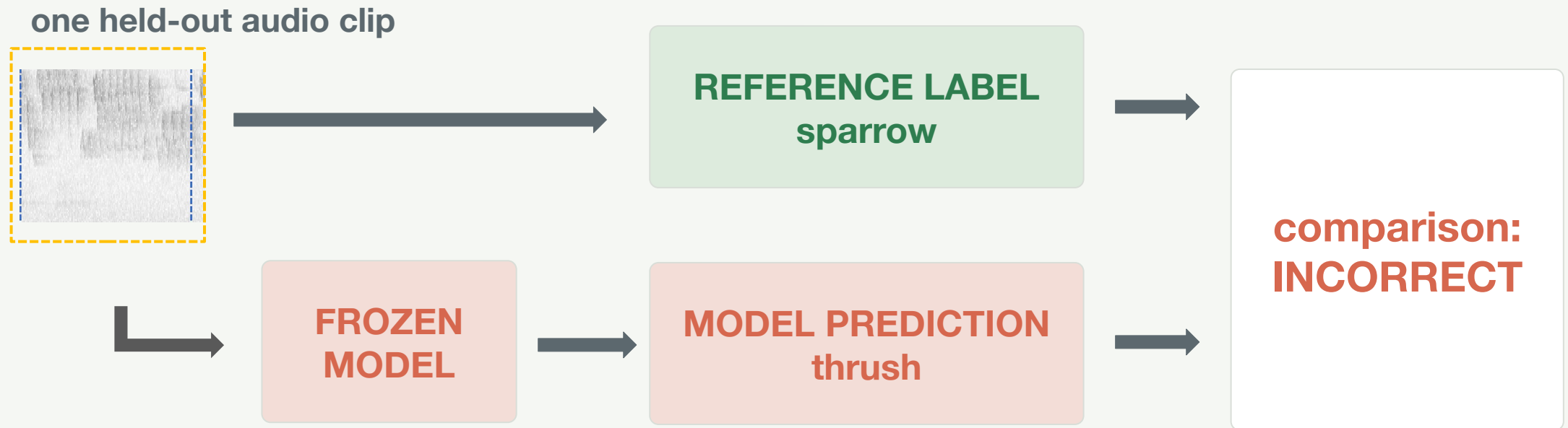


EVALUATION

**How does the FROZEN
system behave
across a defined set of cases?**

**frozen = unchanged model + unchanged
decision rule**

Reference Labels and Predictions

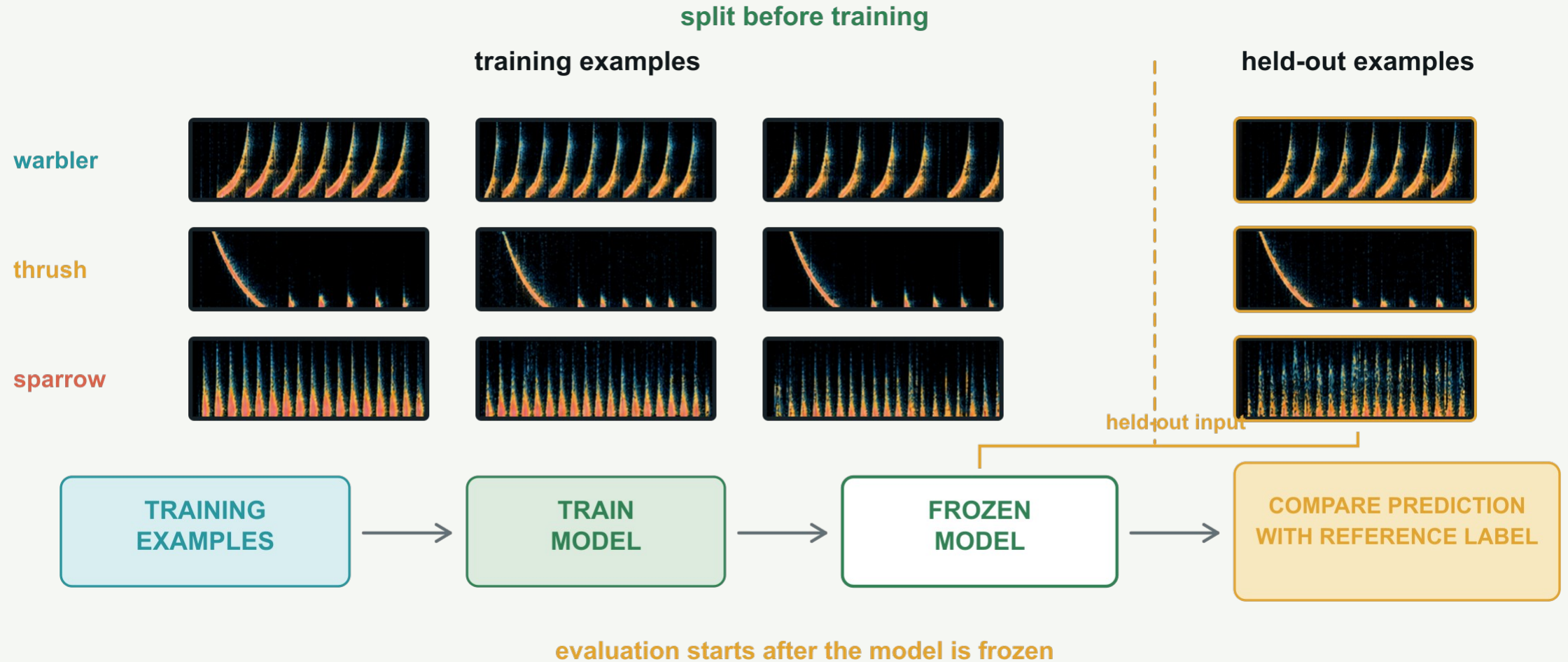


How do we get a REFERENCE LABEL?

Evaluation Cases

test case	ground truth	prediction	correct?
clip 01	warbler	warbler	yes
clip 02	thrush	thrush	yes
clip 03	sparrow	thrush	no
clip 04	other	other	yes
clip 05	thrush	warbler	no
clip 06	sparrow	sparrow	yes
clip 07	warbler	warbler	yes
clip 08	other	other	yes

Held-Out Evaluation Data



Accuracy

$$\frac{6 \text{ correct}}{8 \text{ evaluated}} = 75\% \text{ accuracy}$$

Useful summary. Incomplete description.

Confusion Matrix Construction

		predicted		
		warbler	thrush	sparrow
true	warbler	0	0	0
	thrush	0	0	0
	sparrow	0	+1	0

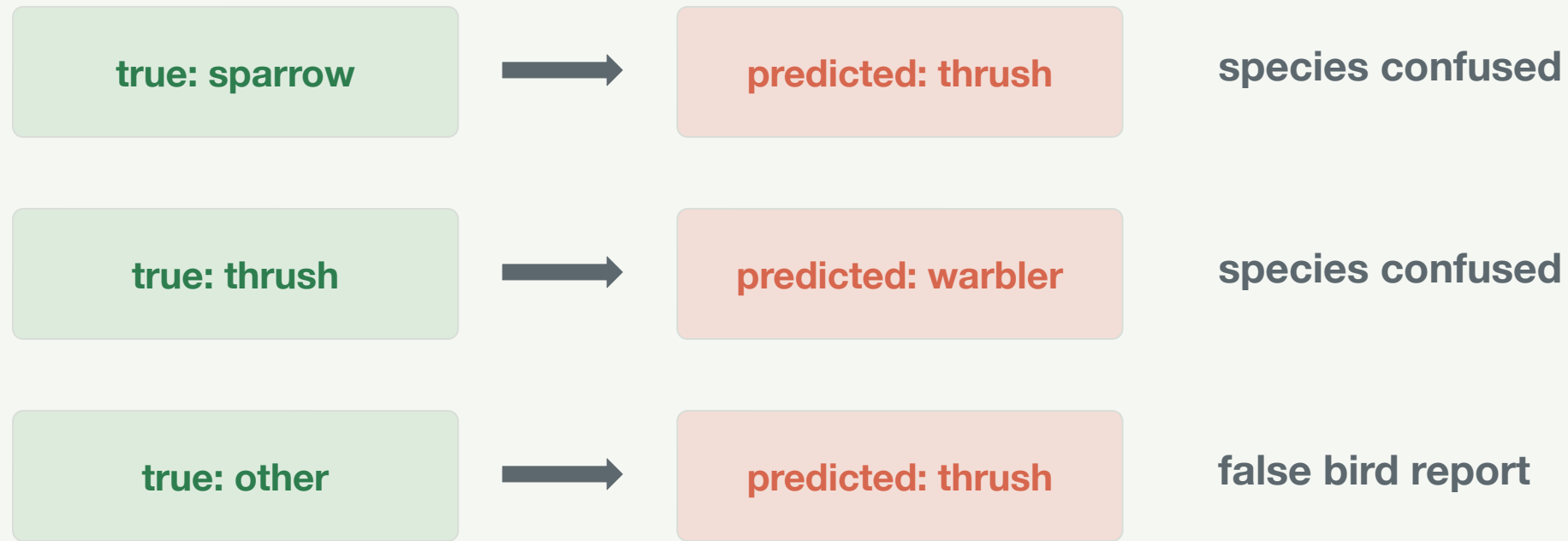
Diagram illustrating Confusion Matrix Construction:

Input: true: sparrow, predicted: thrush

Resulting Confusion Matrix (rows are true labels, columns are predicted labels):

	warbler	thrush	sparrow
warbler	0	0	0
thrush	0	0	0
sparrow	0	+1	0

Classification Error Types



Confusion Matrix

		predicted			
		warbler	thrush	sparrow	other
true	warbler	16	2	1	1
	thrush	2	14	2	2
	sparrow	1	2	16	1
	other	2	2	1	15

Reading a Confusion Matrix

		predicted			
		warbler	thrush	sparrow	other
true	warbler	16	2	1	1
	thrush	2	14	2	2
	sparrow	1	2	16	1
	other	2	2	1	15

READ ACROSS
THE THRUSH
ROW

Where did the real thrush recordings go?

READ
DOWN
THE Other
COLUMN

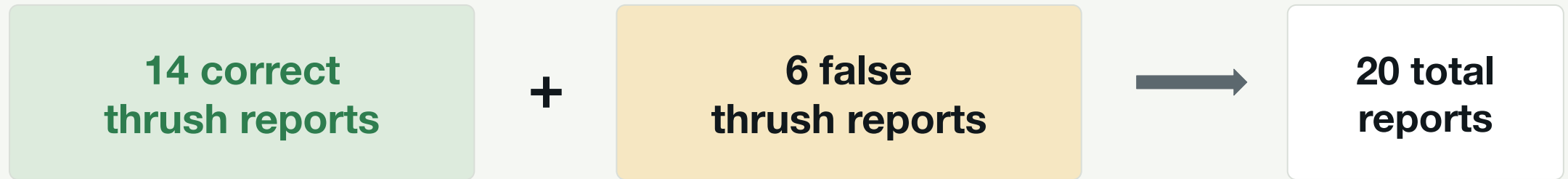
What produced an Other prediction?

One-vs-Rest Confusion Matrix

Target class: THRUSH

	predicted thrush	predicted not thrush
true thrush	TRUE POSITIVE TP = 14	FALSE NEGATIVE FN = 6
true not thrush	FALSE POSITIVE FP = 6	TRUE NEGATIVE TN = 54

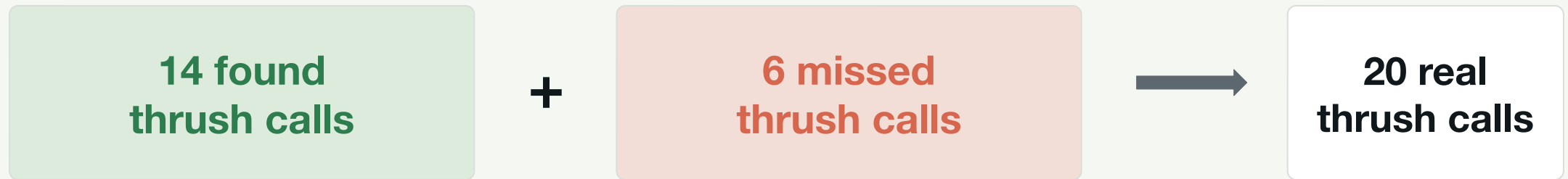
Precision



$$\text{precision} = 14 / (14 + 6) = 0.70$$

Precision measures the quality of accepted positive reports.

Recall



$$\text{recall} = 14 / (14 + 6) = 0.70$$

Recall measures how completely the system finds the positive class.

What can you claim from the confusion matrix?

This classifier recognizes birds in any environment.

On these 80 labeled recordings, thrush precision and recall were both 0.70.

A thrush score of 0.70 means the model is 70% certain.

Specificity and F1 Score

SPECIFICITY

Of all actual negatives,
how many were rejected?

$$TN / (TN + FP)$$

$$54 / (54 + 6) = 0.90$$

F1 SCORE

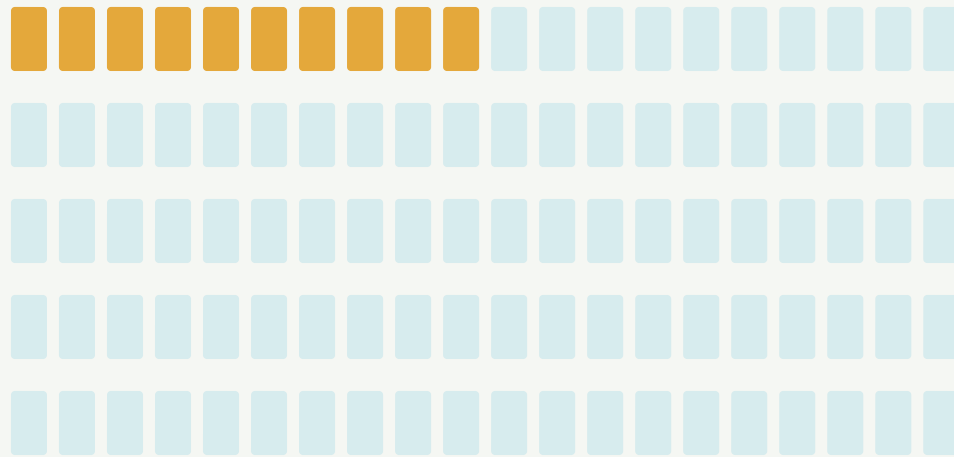
One combined summary
of precision and recall

$$2PR / (P + R)$$

$$F1 = 0.70$$

Class Imbalance

100 field recordings



10 target calls

90 other recordings

Predict "other" every time

90% accuracy

0% target recall